

$P_s = 100 \text{ mW}$ $\rightarrow$ Convert this to dBm

$P_N = 10 \text{ dBm}$ $\rightarrow$ " " " " Watt (mW)

$$B = 20 \text{ MHz}$$

th tell the maximum bitrate
following Shannon's theorem.

$$10 \log_{10} \frac{P_s}{P_N}$$

$$10 \log \frac{100 \text{ mW}}{10 \text{ mW}} = \underline{20 \text{ dBm}}$$

Noise power = 10 dBm

$$\Rightarrow 10 = 10 \log_{10} \frac{P_N}{1 \text{ mW}}$$

$$\Rightarrow P_N = 10 \text{ mW}$$

$$\text{Shan} = B \times \log_2 \left(1 + \frac{P_s}{P_N} \right) = 20 \times \log_2 \left(1 + \frac{P_s}{P_N} \right)$$

dBm

$$\Rightarrow 10 \log \frac{P_x}{1 \text{ mW}}$$

$X \rightarrow S/N \dots$

$$\Rightarrow = 20 \times \log_2 \left(1 + \frac{100}{10} \right)$$

$$= 20 \times \log_2 (1 + 10)$$

$$= 69 \text{ Mbps}$$

